

research was done by Peter Shor. The latest innovation is about generating sufficient redundancy to maintain information integrity. These redundancy methods have negatively introduced logarithmic downsizing on actual logical qubits available for compute, and as of today, we are barely in the range of a few dozen logical qubits,” explains Mahajan.

Subramaniam shares that, “At IBM, we are developing bottom-up approaches to the problem of noisy qubits and incorporating error mitigation techniques to realise this technology’s true potential. Superconducting qubit based systems have made tremendous strides in device performance, from improved coherences to lowered single- and two-qubit gate errors, and high-fidelity mid-circuit measurements and qubit resets. In the future, we plan to bring together advances in device performance, software and error correction to offer users a frictionless experience and the ability to solve tough computational problems.”

In 2021, IBM unveiled Eagle, IBM’s first quantum processor developed and deployed to contain more than 100 operational and connected qubits. “It is the first IBM quantum processor whose scale makes it impossible for a classical computer to reliably simulate,” says Subramaniam.

IBM has the 433-qubit Osprey processor slated for release later this year, and the 1121-qubit Condor processor slated for release next year. In 2024, IBM plans to introduce longer-range quantum communication between chips and create clusters of quantum processors using a long-range coupler for connecting qubit chips through a cryogenic cable of around a metre long. This will be demonstrated by linking together at least three 462-qubit processors, each called Flamingo, into a 1386-qubit system.

IBM’s next quantum processor, Kookaburra, is also slated for release in 2025. It will be a 1386-qubit multichip processor with quantum communication



Quantum Communication Lab (Credit: IIT Madras)

link support for quantum parallelisation. IBM also plans to connect three Kookaburra chips into a 4158-qubit system connected by quantum communication.

Beyond superconducting qubits

Prof. Prabhakar remarks that most people tend to think only of superconducting qubits from companies such as IBM or Google, when considering quantum computers. “There are other hardware platforms also gaining traction, like ion-traps, cold atoms, and photonic cluster states. These are very different approaches and I expect that they will yield application-specific quantum processing units (QPUs) in the near term,” he says.

QCI has developed one such system, which Liscouski describes to us. “Natural quantum states interact freely, influencing and impacting each other as they evolve and change. This natural interaction (or ‘noise’) significantly impacts the accuracy and scale of first generation NISQ computers. These vendors painstakingly create hyper-cooled, vacuum environments to house their computers, but even this doesn’t prevent them from making significant errors, losing information, and suffering from limited scale. Our photonics-based Entropy Quantum Computer (EQC), on the other hand, harnesses the fundamentals of quantum physics to overcome these limitations. It operates on open quantum systems, carefully coupling to an engineered environment, so that the quantum state is collapsed to represent a problem’s desirable solution. In

layman’s terms, where other quantum computers try to avoid noise, our system uses the noise,” he says.

As a result, QCI’s EQC does not require any specialised or costly infrastructure, and needs minimal calibration. It is stable and maintains coherence at room temperature, in any environment. So, there is no restriction to where and how it can be deployed.

Xanadu’s Borealis is also a photon based quantum computer, where a quantum light source, with adjustable brightness, emits trains of up to 288 squeezed-state qubits.

Honeywell Quantum Solutions uses trapped ions to achieve their quantum goals. According to their documentation, their systems ‘trap’ charged ytterbium atoms (ions) with electromagnetic fields so they can be manipulated and encoded with information using microwave signals and lasers. Their latest offering, the System Model H1, promises high-quality quantum operations (fidelities) and longer coherence times than other quantum computing technologies.

Honeywell has collaborated with Cambridge Quantum, which makes AI based quantum software and algorithms, to form Quantinuum, to offer an integrated end-to-end quantum platform that is entirely platform agnostic. Their first product, Quantum Origin, is the world’s first cryptographic key generation platform based on verifiable quantum randomness. They are also developing solutions for chemistry, natural language processing, machine learning, and more.

IonQ, the first quantum computing startup to go public, uses individual atoms as their qubits. According to company literature, “At IonQ, we take a different approach, and use a naturally occurring quantum system: individual atoms. These atoms are the heart of our quantum processing units. We trap them in 3D space, and then use lasers to do everything from initial preparation to final readout.”